

1 **Supplementary Appendix: Null Simulation of Catamenial Epilepsy Classification**

2 This appendix provides additional methodological details, prespecified definitions, quality-control checks, and supplementary results
3 for the Paper 1 null simulation. It is intended as online supplementary material accompanying the main Neurology submission draft.

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12 **A1. Simulation design and reproducibility**

13 The analysis used a deterministic master seed of 20260505. The full run simulated 100,000 synthetic participants, processed in
14 chunks, with 50,000 in the healthy ovulatory cohort and 50,000 in the broad population cohort. The 36-month diaries were generated
15 independently for seizures and menstrual/hormone cycles, followed by a participant-specific random circular shift before merging.

16 Daily audit data were saved for a 1% random sample. Participant-level and window-level outputs were written as parquet and CSV
17 files, and every output artifact was recorded in a machine-readable manifest with file size and SHA-256 checksum.

18 **A2. Cohort definitions and diary generation**

19 The healthy ovulatory cohort targeted adult ovulatory cycling, age 18-45 years, with simulator medical modifiers disabled where
20 exposed. The broad population cohort targeted a broader menstruating-age range of 13.0-54.9 years and allowed anovulation and
21 irregularity. Because the hormone simulator exposed medical-factor knobs but not a natural prevalence sampler, PCOS, peri-
22 menarche, perimenopause, and dysmenorrhea were sampled from config.yaml rates.

23 Supplementary Table 1. Cohort construction parameters.

Parameter	Healthy ovulatory	Broad population
Target sample size	50,000	50,000
Age range	18.0-45.0 years	13.0-54.9 years
Ovulatory-cycle handling	Ovulation probability set to 1.0	Natural ovulatory/anovulatory behavior allowed
PCOS rate	Disabled	10%
Peri-menarche rate	Disabled	60% when age <20 years
Perimenopause rate	Disabled	55% when age >=45 years
Dysmenorrhea rate	Disabled	12%

25 A3. Phase labeling and catamenial epilepsy definitions

26 Herzog-style phases were assigned on the full diary before window subsetting. For a complete cycle of length L , day 1 was the onset
 27 of menstrual flow and backward day was defined as $d-(L+1)$. Menstrual phase included days 1-3 and backward days -3 to -1; follicular
 28 phase included days 4-9; ovulatory phase included days ≥ 10 and backward days ≤ -13 ; luteal phase included backward days -12 to
 29 -4.

30 Supplementary Table 2. Implemented catamenial epilepsy definitions.

Definition	Primary rule	Main analyzability condition
Exact Herzog 2004	Three-cycle legacy rule; any CE if at least 2 of 3 cycles show any C1, C2, or C3 pattern.	Only complete 3-cycle windows with strict 23-35 day cycles.
Windowed Herzog thresholds	Window-pooled average daily seizure-frequency ratios for C1, C2, and applicable C3.	Required phases observed; undefined ratios marked indeterminate.
Minimum-data rule	Windowed Herzog thresholds after applying minimum duration and seizure-burden criteria.	At least 4 months or 6 complete cycles, plus at least 4 seizure days.
Cycle reproducibility	At least two-thirds of eligible cycles positive for the same pattern and pooled ratio passes threshold.	At least 6 complete cycles for primary rule; 12 cycles for sensitivity.
Negative-binomial regression	One-sided Holm-adjusted tests for menstrual and ovulatory enrichment with rate-ratio thresholds.	Minimum-data rule passed; Poisson robust fallback recorded when needed.
Historical rules	Assumption-based operationalizations of Newmark-Perry, Duncan 1993, Herzog 1997 twofold, and Reddy 2007 any-phase rules.	Reported separately and flagged as assumption based.

32 **A4. Observation windows and analyzability rules**

33 Calendar windows used one random valid start day per participant for 1, 3, 4, 6, and 12 months. Cycle windows used one random
34 valid start among complete cycles for 3, 6, and 12 cycles. The full-window analysis used the entire 36-month diary. Windows were
35 marked indeterminate, not negative, when required phases or denominator information were unavailable.

36 **A5. Statistical analysis and edge-case handling**

37 All ratio definitions used average daily seizure frequency rather than raw counts. If a comparator phase had observed days but zero
38 seizures, a positive numerator produced an infinite ratio and counted positive; zero numerator and zero comparator produced an
39 undefined ratio and an indeterminate result. Wilson 95% confidence intervals were used for simple proportions. Study-level Monte
40 Carlo sampled 10,000 studies of 30 participants per cohort for the 3-month window.

41 **A6. Supplementary results**

42 Supplementary Table 3. Window-length sensitivity for core definitions.

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
broad population	1 month	Cycle reproducibility rule	0	0	NA	100.0%
broad population	1 month	Exact Herzog 2004, any pattern	0	0	NA	100.0%
broad population	1 month	Negative-binomial regression	0	0	NA	100.0%
broad population	1 month	Windowed Herzog thresholds	38,083	19,748	51.9%	23.8%
broad population	1 month	Windowed Herzog with minimum data	0	0	NA	100.0%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
broad population	3 months	Cycle reproducibility rule	0	0	NA	100.0%
broad population	3 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
broad population	3 months	Negative-binomial regression	0	0	NA	100.0%
broad population	3 months	Windowed Herzog thresholds	45,297	23,184	51.2%	9.4%
broad population	3 months	Windowed Herzog with minimum data	0	0	NA	100.0%
broad population	4 months	Cycle reproducibility rule	0	0	NA	100.0%
broad population	4 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
broad population	4 months	Negative-binomial regression	37,529	1,180	3.1%	24.9%
broad population	4 months	Windowed Herzog thresholds	46,369	23,060	49.7%	7.3%
broad population	4 months	Windowed Herzog with minimum data	37,530	17,825	47.5%	24.9%
broad population	6 months	Cycle reproducibility rule	5,189	409	7.9%	89.6%
broad population	6 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
broad population	6 months	Negative-binomial regression	41,240	1,862	4.5%	17.5%
broad population	6 months	Windowed Herzog thresholds	47,490	22,546	47.5%	5.0%
broad population	6 months	Windowed Herzog with minimum data	41,240	18,769	45.5%	17.5%
broad population	12 months	Cycle reproducibility rule	20,438	1,279	6.3%	59.1%
broad population	12 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
broad population	12 months	Negative-binomial regression	45,303	2,122	4.7%	9.4%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
broad population	12 months	Windowed Herzog thresholds	48,710	20,543	42.2%	2.6%
broad population	12 months	Windowed Herzog with minimum data	45,303	18,437	40.7%	9.4%
broad population	3 cycles	Cycle reproducibility rule	0	0	NA	100.0%
broad population	3 cycles	Exact Herzog 2004, any pattern	16,673	8,602	51.6%	66.7%
broad population	3 cycles	Negative-binomial regression	1,947	50	2.6%	96.1%
broad population	3 cycles	Windowed Herzog thresholds	45,349	22,907	50.5%	9.3%
broad population	3 cycles	Windowed Herzog with minimum data	1,947	1,140	58.6%	96.1%
broad population	6 cycles	Cycle reproducibility rule	25,842	2,006	7.8%	48.3%
broad population	6 cycles	Exact Herzog 2004, any pattern	0	0	NA	100.0%
broad population	6 cycles	Negative-binomial regression	41,326	1,910	4.6%	17.3%
broad population	6 cycles	Windowed Herzog thresholds	47,488	22,375	47.1%	5.0%
broad population	6 cycles	Windowed Herzog with minimum data	41,326	18,672	45.2%	17.3%
broad population	12 cycles	Cycle reproducibility rule	20,074	1,854	9.2%	59.9%
broad population	12 cycles	Exact Herzog 2004, any pattern	0	0	NA	100.0%
broad population	12 cycles	Negative-binomial regression	45,332	2,153	4.7%	9.3%
broad population	12 cycles	Windowed Herzog thresholds	48,706	20,061	41.2%	2.6%
broad population	12 cycles	Windowed Herzog with minimum data	45,332	18,002	39.7%	9.3%
broad population	36-month full diary	Cycle reproducibility rule	13,364	1,605	12.0%	73.3%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
broad population	36-month full diary	Exact Herzog 2004, any pattern	0	0	NA	100.0%
broad population	36-month full diary	Negative-binomial regression	48,333	2,032	4.2%	3.3%
broad population	36-month full diary	Windowed Herzog thresholds	49,587	17,983	36.3%	0.8%
broad population	36-month full diary	Windowed Herzog with minimum data	48,333	17,206	35.6%	3.3%
healthy ovulatory	1 month	Cycle reproducibility rule	0	0	NA	100.0%
healthy ovulatory	1 month	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	1 month	Negative-binomial regression	0	0	NA	100.0%
healthy ovulatory	1 month	Windowed Herzog thresholds	38,234	18,249	47.7%	23.5%
healthy ovulatory	1 month	Windowed Herzog with minimum data	0	0	NA	100.0%
healthy ovulatory	3 months	Cycle reproducibility rule	0	0	NA	100.0%
healthy ovulatory	3 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	3 months	Negative-binomial regression	0	0	NA	100.0%
healthy ovulatory	3 months	Windowed Herzog thresholds	45,280	18,696	41.3%	9.4%
healthy ovulatory	3 months	Windowed Herzog with minimum data	0	0	NA	100.0%
healthy ovulatory	4 months	Cycle reproducibility rule	0	0	NA	100.0%
healthy ovulatory	4 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	4 months	Negative-binomial regression	37,641	1,236	3.3%	24.7%
healthy ovulatory	4 months	Windowed Herzog	46,381	17,894	38.6%	7.2%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
		thresholds				
healthy ovulatory	4 months	Windowed Herzog with minimum data	37,641	13,501	35.9%	24.7%
healthy ovulatory	6 months	Cycle reproducibility rule	6,706	704	10.5%	86.6%
healthy ovulatory	6 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	6 months	Negative-binomial regression	41,136	1,881	4.6%	17.7%
healthy ovulatory	6 months	Windowed Herzog thresholds	47,489	16,185	34.1%	5.0%
healthy ovulatory	6 months	Windowed Herzog with minimum data	41,136	12,985	31.6%	17.7%
healthy ovulatory	12 months	Cycle reproducibility rule	15,964	246	1.5%	68.1%
healthy ovulatory	12 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	12 months	Negative-binomial regression	45,266	2,207	4.9%	9.5%
healthy ovulatory	12 months	Windowed Herzog thresholds	48,775	12,419	25.5%	2.4%
healthy ovulatory	12 months	Windowed Herzog with minimum data	45,266	10,661	23.6%	9.5%
healthy ovulatory	3 cycles	Cycle reproducibility rule	0	0	NA	100.0%
healthy ovulatory	3 cycles	Exact Herzog 2004, any pattern	23,101	11,616	50.3%	53.8%
healthy ovulatory	3 cycles	Negative-binomial regression	17	1	5.9%	100.0%
healthy ovulatory	3 cycles	Windowed Herzog thresholds	45,122	18,831	41.7%	9.8%
healthy ovulatory	3 cycles	Windowed Herzog with minimum data	17	12	70.6%	100.0%
healthy ovulatory	6 cycles	Cycle reproducibility rule	22,591	2,742	12.1%	54.8%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
healthy ovulatory	6 cycles	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	6 cycles	Negative-binomial regression	40,861	1,860	4.6%	18.3%
healthy ovulatory	6 cycles	Windowed Herzog thresholds	47,394	16,299	34.4%	5.2%
healthy ovulatory	6 cycles	Windowed Herzog with minimum data	40,861	12,925	31.6%	18.3%
healthy ovulatory	12 cycles	Cycle reproducibility rule	15,706	355	2.3%	68.6%
healthy ovulatory	12 cycles	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	12 cycles	Negative-binomial regression	45,088	2,155	4.8%	9.8%
healthy ovulatory	12 cycles	Windowed Herzog thresholds	48,682	12,364	25.4%	2.6%
healthy ovulatory	12 cycles	Windowed Herzog with minimum data	45,088	10,544	23.4%	9.8%
healthy ovulatory	36-month full diary	Cycle reproducibility rule	6,132	1	0.0%	87.7%
healthy ovulatory	36-month full diary	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	36-month full diary	Negative-binomial regression	48,359	1,966	4.1%	3.3%
healthy ovulatory	36-month full diary	Windowed Herzog thresholds	49,605	5,576	11.2%	0.8%
healthy ovulatory	36-month full diary	Windowed Herzog with minimum data	48,359	4,940	10.2%	3.3%

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44 Supplementary Table 4. Study-level Monte Carlo results.

Cohort	Definition	Denominator	Mean apparent prevalence	2.5th-97.5th percentiles	P >=39.1%	P >=44.2%
broad population	Cycle reproducibility rule	All participants	0.0%	0.0%-0.0%	0.0%	0.0%
broad population	Cycle reproducibility rule	Classifiable participants only	NA	NA-NA	NA	NA
broad population	Negative-binomial regression	All participants	0.0%	0.0%-0.0%	0.0%	0.0%
broad population	Negative-binomial regression	Classifiable participants only	NA	NA-NA	NA	NA
broad population	Windowed Herzog thresholds	All participants	46.0%	30.0%-63.3%	79.9%	54.5%
broad population	Windowed Herzog thresholds	Classifiable participants only	50.8%	32.0%-69.2%	89.4%	75.9%
broad population	Windowed Herzog with minimum data	All participants	0.0%	0.0%-0.0%	0.0%	0.0%
broad population	Windowed Herzog with minimum data	Classifiable participants only	NA	NA-NA	NA	NA
healthy ovulatory	Cycle reproducibility rule	All participants	0.0%	0.0%-0.0%	0.0%	0.0%
healthy ovulatory	Cycle reproducibility rule	Classifiable participants only	NA	NA-NA	NA	NA
healthy ovulatory	Negative-binomial regression	All participants	0.0%	0.0%-0.0%	0.0%	0.0%
healthy ovulatory	Negative-binomial regression	Classifiable participants only	NA	NA-NA	NA	NA
healthy ovulatory	Windowed Herzog thresholds	All participants	37.5%	20.0%-53.3%	45.9%	19.7%
healthy ovulatory	Windowed Herzog thresholds	Classifiable participants only	41.4%	23.1%-60.0%	60.8%	39.5%
healthy ovulatory	Windowed Herzog with minimum data	All participants	0.0%	0.0%-0.0%	0.0%	0.0%
healthy ovulatory	Windowed Herzog with minimum data	Classifiable participants only	NA	NA-NA	NA	NA

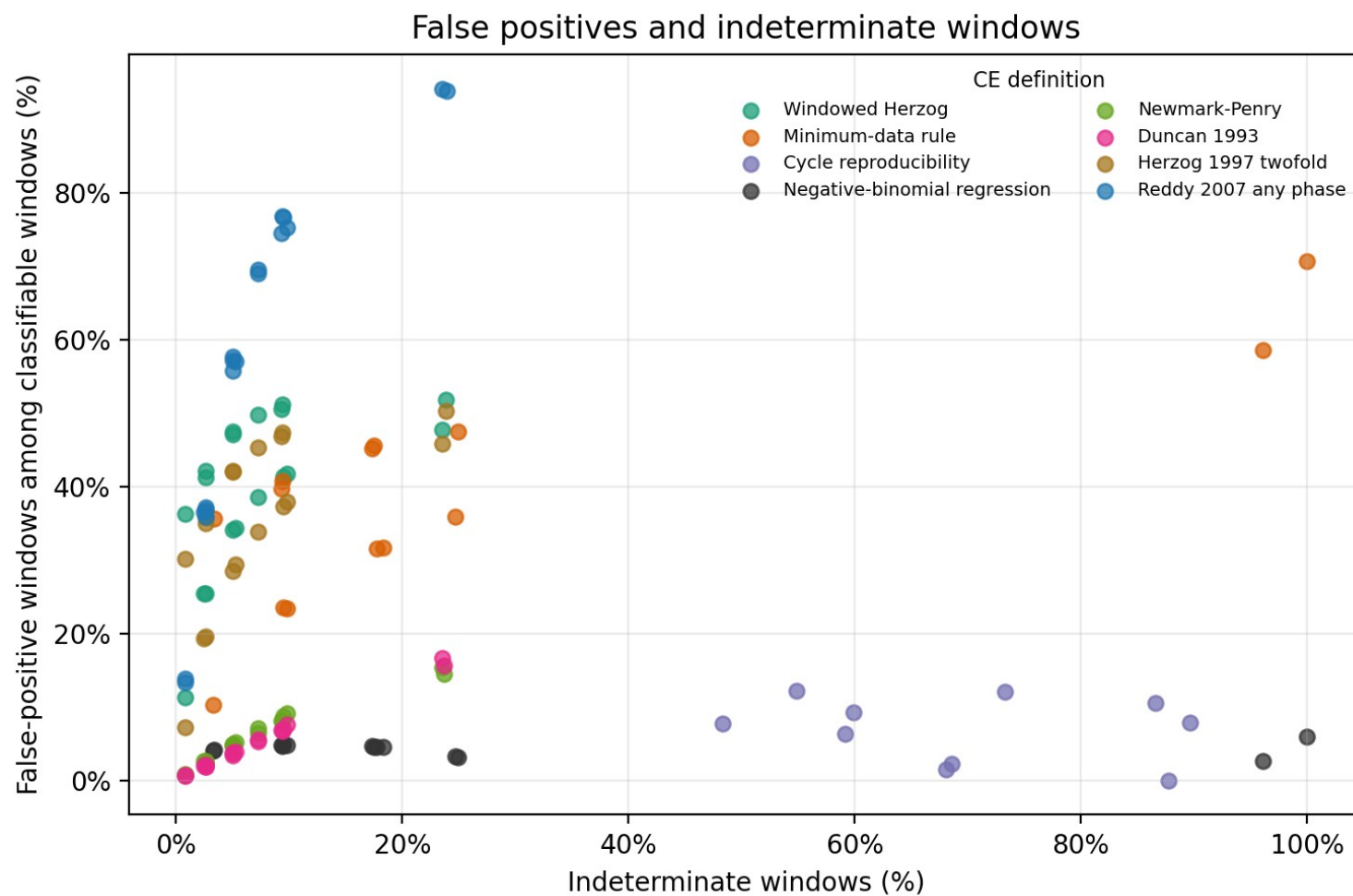
46 Supplementary Table 5. Assumption-based historical definitions.

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
broad population	3 months	Duncan 1993 ten-day rule	45,297	3,018	6.7% (6.4%-6.9%)	9.4%
broad population	3 months	Herzog 1997 twofold rule	45,297	21,417	47.3% (46.8%-47.7%)	9.4%
broad population	3 months	Newmark-Penry perimenstrual rule	45,297	3,696	8.2% (7.9%-8.4%)	9.4%
broad population	3 months	Newmark-Penry two-thirds sensitivity	45,297	1,733	3.8% (3.7%-4.0%)	9.4%
broad population	3 months	Reddy 2007 any-phase twofold rule	45,297	34,750	76.7% (76.3%-77.1%)	9.4%
broad population	36-month full diary	Duncan 1993 ten-day rule	49,587	311	0.6% (0.6%-0.7%)	0.8%
broad population	36-month full diary	Herzog 1997 twofold rule	49,587	14,914	30.1% (29.7%-30.5%)	0.8%
broad population	36-month full diary	Newmark-Penry perimenstrual rule	49,587	378	0.8% (0.7%-0.8%)	0.8%
broad population	36-month full diary	Newmark-Penry two-thirds sensitivity	49,587	175	0.4% (0.3%-0.4%)	0.8%
broad population	36-month full diary	Reddy 2007 any-phase twofold rule	49,587	6,854	13.8% (13.5%-14.1%)	0.8%
healthy ovulatory	3 months	Duncan 1993 ten-day rule	45,280	3,163	7.0% (6.8%-7.2%)	9.4%
healthy ovulatory	3 months	Herzog 1997 twofold rule	45,280	16,862	37.2% (36.8%-37.7%)	9.4%
healthy ovulatory	3 months	Newmark-Penry perimenstrual rule	45,280	3,939	8.7% (8.4%-9.0%)	9.4%
healthy ovulatory	3 months	Newmark-Penry two-thirds sensitivity	45,280	1,870	4.1% (4.0%-4.3%)	9.4%
healthy ovulatory	3 months	Reddy 2007 any-phase twofold rule	45,280	34,681	76.6% (76.2%-77.0%)	9.4%
healthy ovulatory	36-month full diary	Duncan 1993 ten-day rule	49,605	361	0.7% (0.7%-0.8%)	0.8%
healthy ovulatory	36-month full diary	Herzog 1997 twofold rule	49,605	3,603	7.3% (7.0%-7.5%)	0.8%
healthy ovulatory	36-month full diary	Newmark-Penry perimenstrual rule	49,605	427	0.9% (0.8%-0.9%)	0.8%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
healthy ovulatory	36-month full diary	Newmark-Penry two-thirds sensitivity	49,605	170	0.3% (0.3%-0.4%)	0.8%
healthy ovulatory	36-month full diary	Reddy 2007 any-phase twofold rule	49,605	6,599	13.3% (13.0%-13.6%)	0.8%

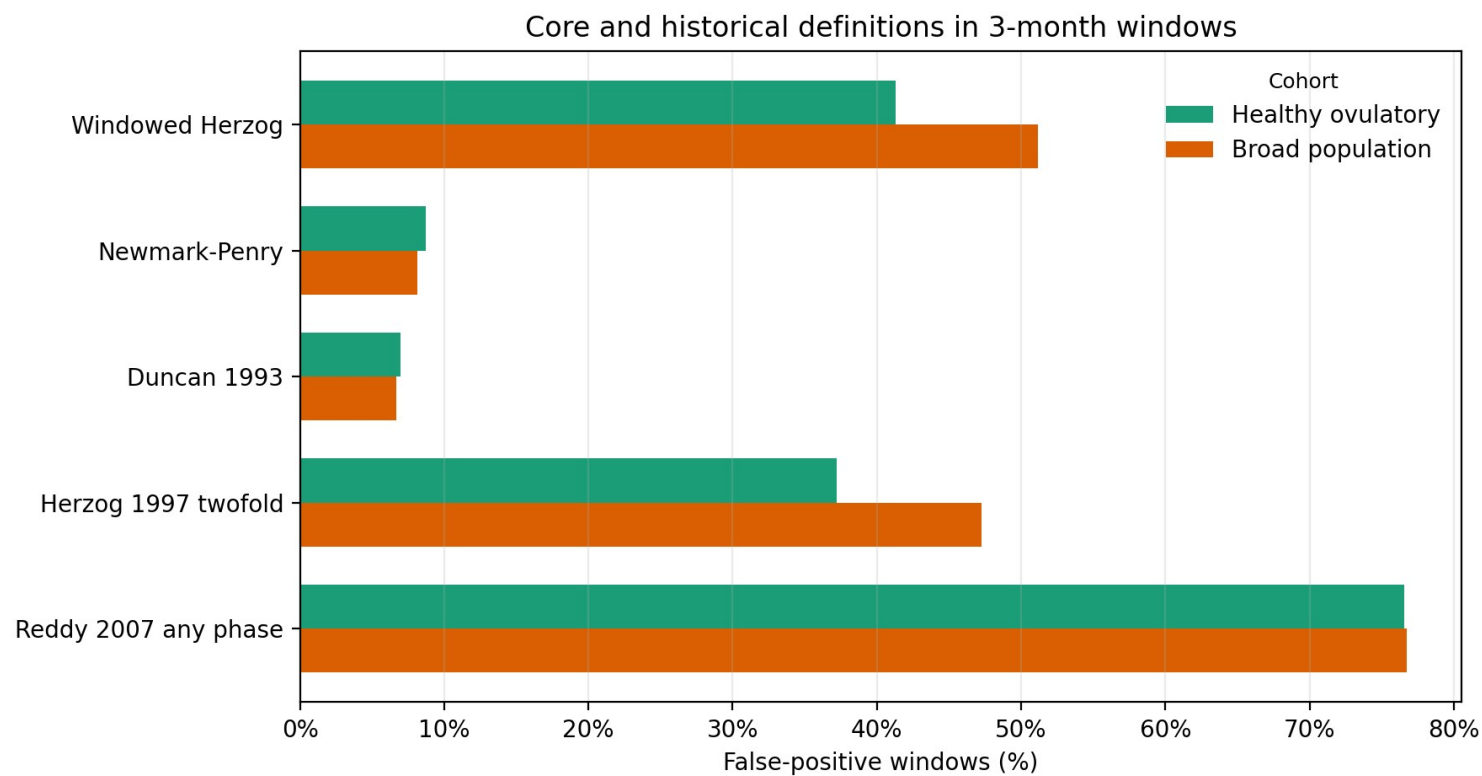
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48 Supplementary Figure 1. False-positive and indeterminate frontier.



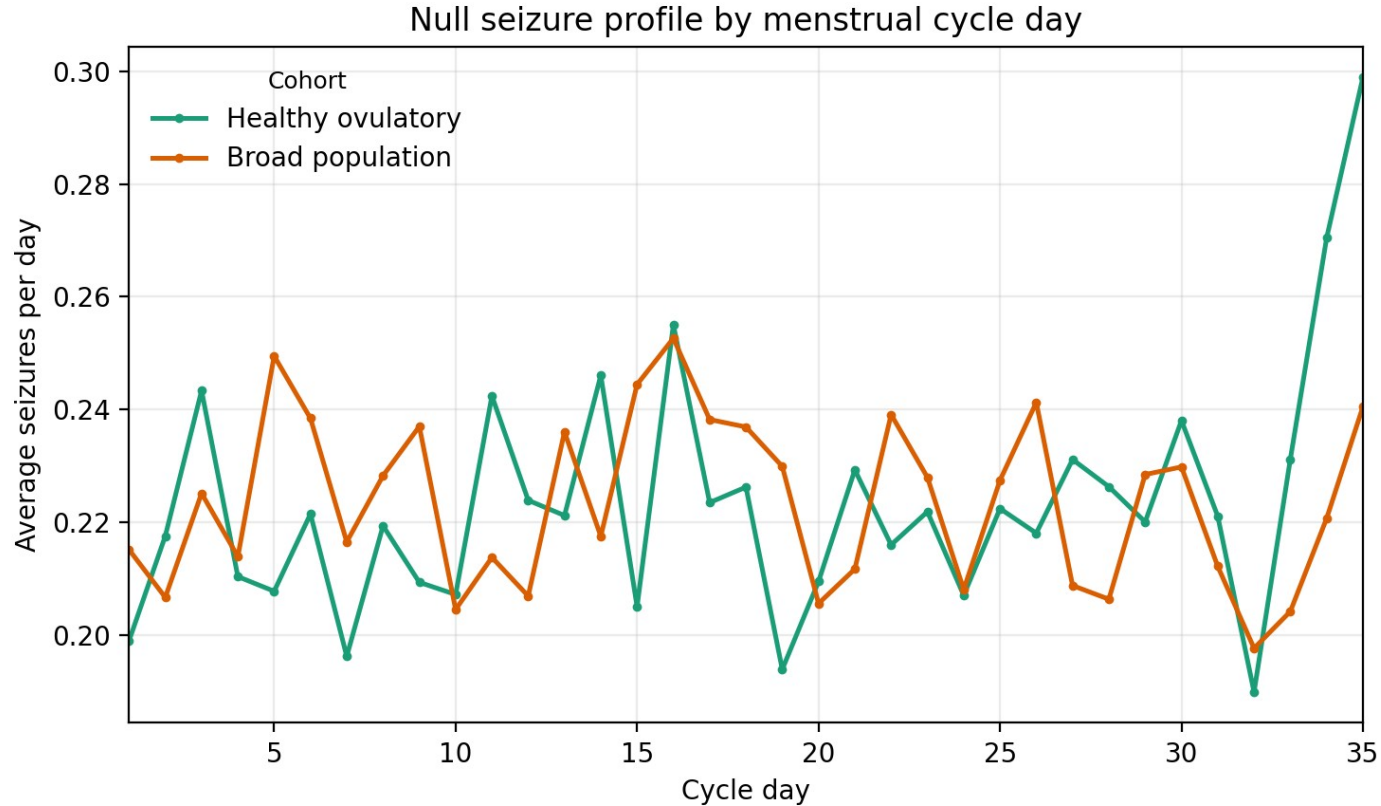
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50 Supplementary Figure 2. Historical and core definitions in 3-month windows.



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52 Supplementary Figure 3. Null seizure profile by menstrual cycle day.



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54 **A7. Output provenance and assumptions**

- 55 • Definition D uses a participant-full-diary method-of-moments negative-binomial alpha recorded in d_alpha; Poisson robust
- 56 fallback is recorded in d_reason when statsmodels negative-binomial fitting fails.
- 57 • Healthy ovulatory cohort used hormone_cycler build_patient_profile/render_cycle with ovulation_probability set to 1.0 because
- 58 simulate_diary did not expose a public force-ovulation knob.

- 59 • Historical definitions H1-H4 are assumption-based operationalizations and are flagged in summary outputs.
- 60 • Study-level Monte Carlo samples each selected participant from a deterministic pool of precomputed random valid 3-month
- 61 windows to avoid retaining all daily diaries in memory.
- 62 • The hormone simulator exposes medical-factor knobs but no natural prevalence sampler; broad population medical factors were
- 63 sampled from config.yaml rates.